The factor u of the P18 profile: a critically damped approach to the isothermal sphere, and a burst experiment

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companion note to *From a primordial fluctuation to an idealised rotation curve* (pf2irc), Sects. 2.1–2.3

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Abstract

The P18 profile is the singular isothermal sphere (SIS) multiplied by u^2, with $u(x) = 1 + (x-1)e^{-x}$ and $x = R/r_c$. We show that u is the unique solution of the linear equation $u'' + 2u' + u = 1$ with $u(0) = 0$ and $u'(0) = 2$: the approach of the profile to the SIS is that of a *critically damped* oscillator, with the radius in the role of time and the core radius as damping length. The regular solutions form a one-parameter family $u_b = 1 + (bx-1)e^{-x}$. For it we prove that the density crosses that of the SIS at $x = 1/b$, overshoots it if and only if $b > 0$, and carries the mass excess $c(b) = b^2/4 + 3b/2 - 3/2$, which is positive if and only if $b > \sqrt{15} - 3$; P18 is the member $b = 1$, with $c = \frac{1}{4}$, the only one whose damping length equals the core radius defined by the central density. The non-singular isothermal sphere, by contrast, approaches the SIS *underdamped* in $\ln x$, as $x^{-1/2}\cos(\frac{\sqrt{7}}{2}\ln x + \varphi)$, without a length scale. The double characteristic root has a simple reading: two identical exponential relaxations in series. To test whether outflows from central starbursts, decelerated by the infall, could produce such a single overshoot, we add to the paper's self-similar collisionless shells a central baryonic mass that is repeatedly blown out and falls back. The bursts flatten the cusp (density slope from -2.5 to between -0.8 and -1.8) and rearrange the dark matter in a P18-like pattern, a rise, a single peak and a zone of excess, but always on the scale of the accretion front, not of the core, even when the gas is confined to two core radii. On nearly radial orbits the displaced matter is deposited near its apocentres, where the infall turns it round; an overshoot at the core would require the orbits there to become isotropic.

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1 The question

In the paper the total matter of a late-type dwarf galaxy follows the P18 profile,

$$\rho(R) = \frac{v^2 u(x)^2}{4\pi G R^2} = \rho_{\rm SIS}(R)\, u(x)^2, \qquad u(x) = 1 + (x-1)\, e^{-x}, \qquad x = \frac{R}{r_c}, \tag{1}$$

with $\rho(0) = v^2/(\pi G r_c^2)$. All the information about the core is in the factor u: for $u \equiv 1$ the profile is the singular isothermal sphere (SIS). The form of u is empirical. This note asks two questions:

1. What is the mathematical structure of u? (Sects. 2–5)
2. Can a simple physical process, outflows from early central starbursts slowed down by the continuing infall, produce a deviation from the SIS of this kind? (Sect. 6)

Throughout, $D \equiv \mathrm{d}/\mathrm{d}x$, $g \equiv u - 1$, and $m(x) = \int_0^x u^2\, \mathrm{d}x'$ is the enclosed mass in units of $v^2 r_c/G$.

2 The differential equation

Proposition 1. *The function $u(x) = 1 + (x-1)e^{-x}$ is the unique solution on $[0, \infty)$ of*

$$u'' + 2u' + u = 1, \qquad u(0) = 0, \quad u'(0) = 2. \tag{2}$$

Equivalently, $g = u - 1$ satisfies $(D+1)^2 g = 0$ with $g(0) = -1$, $g'(0) = 2$.

Proof. With $g = (x-1)e^{-x}$ one finds $g' = (2-x)e^{-x}$ and $g'' = (x-3)e^{-x}$, so $g'' + 2g' + g = (x - 3 + 4 - 2x + x - 1)e^{-x} = 0$; hence $u = 1 + g$ satisfies $u'' + 2u' + u = 1$. Further $u(0) = 1 - 1 = 0$ and $u'(0) = g'(0) = 2$. The equation is linear with constant coefficients, so by the existence and uniqueness theorem for linear initial-value problems its solution with given $u(0)$ and $u'(0)$ is unique. □

The characteristic polynomial of the homogeneous equation is $s^2 + 2s + 1 = (s+1)^2$, with the double root $s = -1$. In the language of the damped oscillator, $\ddot{q} + 2\gamma\dot{q} + \omega_0^2 q = 0$ with $\gamma = \omega_0 = 1$: the damping is exactly *critical*. The variable x plays the role of time, the deviation g from the isothermal value $u = 1$ that of the displacement, and the core radius that of the damping length.

The general solution and the regular family. Every solution of $u'' + 2u' + u = 1$ has the form $u = 1 + (A + Bx)e^{-x}$. Since $\rho/\rho_{\rm SIS} = u^2$ and $\rho_{\rm SIS} \propto x^{-2}$, the central density is finite if and only if $u(0) = 0$, i.e. $A = -1$. The *regular* solutions therefore form the one-parameter family

$$u_b(x) = 1 + (bx - 1)\, e^{-x}, \qquad u_b'(0) = b + 1, \tag{3}$$

and P18 is the member $b = 1$.

Damping length and core radius. If the equation is written for the dimensional radius, $\lambda^2 u'' + 2\lambda u' + u = 1$ with derivatives with respect to R, its solutions are $u_b(R/\lambda)$: λ is the damping length. Near the centre $u_b \simeq (b+1)R/\lambda$, so $\rho(0) = v^2(b+1)^2/(4\pi G\lambda^2)$. Defining the core radius, as in the paper, by $\rho(0) = v^2/(\pi G r_c^2)$ gives

$$r_c = \frac{2\lambda}{b+1}. \tag{4}$$

Corollary 1. *P18 ($b = 1$) is the only regular member whose damping length equals the core radius defined by the central density.*

3 Properties of the regular family

Proposition 2. *For u_b of Eq.* (3)*:*

(i) Crossing. *For $b > 0$, $u_b < 1$ for $x < 1/b$ and $u_b > 1$ for $x > 1/b$: the density crosses that of the SIS exactly at $x = 1/b$. For $b \le 0$, $u_b < 1$ everywhere.*

(ii) Overshoot. *u_b exceeds 1 somewhere if and only if $b > 0$; it then has a single maximum, $u_{\max} = 1 + b\,e^{-(b+1)/b}$ at $x = (b+1)/b$, and approaches 1 monotonically beyond.*

(iii) Mass excess. *$m(x) \to x + c(b)$ for $x \to \infty$, with*

$$c(b) = \int_0^\infty (u_b^2 - 1)\,\mathrm{d}x = \frac{b^2}{4} + \frac{3b}{2} - \frac{3}{2}, \tag{5}$$

which is positive if and only if $b > \sqrt{15} - 3 \simeq 0.873$.

(iv) Centre. *$u_b = (b+1)x - (b+\frac{1}{2})x^2 + O(x^3)$, so $\rho/\rho(0) = 1 - \frac{2b+1}{b+1}\,x + O(x^2)$: the density falls linearly with radius unless $b = -\frac{1}{2}$.*

Proof. (i) $u_b - 1 = (bx-1)e^{-x}$ has the sign of $bx - 1$. (ii) For $b \le 0$, $bx - 1 < 0$ for all $x \ge 0$. For $b > 0$, $u_b' = (b+1-bx)e^{-x}$ vanishes only at $x = (b+1)/b$, where u_b is maximal; inserting gives $u_{\max}$; beyond, $u_b' < 0$ and $u_b \to 1$. (iii) With $g = (bx-1)e^{-x}$, $u^2 - 1 = 2g + g^2$, and with $\int_0^\infty x^n e^{-x}\mathrm{d}x = n!$ and $\int_0^\infty x^n e^{-2x}\mathrm{d}x = n!/2^{n+1}$, $\int_0^\infty 2g\,\mathrm{d}x = 2(b-1)$ and $\int_0^\infty g^2\,\mathrm{d}x = b^2 \cdot \frac{1}{4} - 2b \cdot \frac{1}{4} + \frac{1}{2}$. The sum is Eq. (5). Its positive root is $b = -3 + \sqrt{15}$. Since $m' = u^2 \to 1$ with an exponentially small difference, $m(x) - x \to c(b)$. (iv) Expanding $e^{-x} = 1 - x + \frac{1}{2}x^2 + O(x^3)$ in $1 + (bx-1)e^{-x}$ gives the series, and $\rho/\rho(0) = [u_b/((b+1)x)]^2$. □

For P18 ($b = 1$): the density crosses the SIS at $x = 1$, i.e. at $R = r_c$; the maximum is $u = 1 + e^{-2} = 1.135$ at $x = 2$ ($u^2 = 1.289$); the mass excess is $c = \frac{1}{4}$; and $\rho/\rho(0) = 1 - \frac{3}{2}x + \ldots$ near the centre (Fig. 1a). The two properties are distinct: an overshoot exists for every $b > 0$, but a positive mass excess needs $b > 0.873$, since the deficit inside the crossing must be outweighed by the excess outside it.

The displaced mass. Define $F(x) = x - m(x)$, the mass that P18 lacks inside x compared with the SIS. Since $m' = u^2$,

$$u^2 = 1 - F'(x)\,: \tag{6}$$

F is the mass carried outward through the sphere of radius x, relative to the SIS. For P18, $F(0) = 0$, F is maximal where $u = 1$, at $x = 1$, with $F(1) = 1 - m(1) = 0.520$; it vanishes at $x = 3.29$ and tends to $-c = -0.25$. So $0.520\,v^2 r_c/G$ is taken out of the core and $\int_1^\infty (u^2 - 1)\,\mathrm{d}x = c + 1 - m(1) = 0.770\,v^2 r_c/G$ is added outside it. (Numbers: `ode_checks.py`.)

4 Comparison with the isothermal sphere

A fully thermalised self-gravitating gas, the non-singular isothermal sphere, also has a finite central density and tends to the SIS. With $\rho = \rho_0 e^{-\psi}$ and $\xi = R/r_0$, $r_0^2 = \sigma^2/(4\pi G\rho_0)$, it obeys

$$\psi'' + \frac{2}{\xi}\,\psi' = e^{-\psi}, \qquad \psi(0) = \psi'(0) = 0, \tag{7}$$

whose singular solution $e^{-\psi_s} = 2/\xi^2$ is the SIS. Writing $\psi = \psi_s + \delta$ and linearising, $e^{-\psi} \simeq (2/\xi^2)(1 - \delta)$, so $\delta'' + (2/\xi)\delta' + (2/\xi^2)\delta = 0$. In the variable $s = \ln\xi$ this is

$$\delta_{ss} + \delta_s + 2\delta = 0, \qquad \text{roots } s = -\tfrac{1}{2} \pm \tfrac{\sqrt{7}}{2}\,i, \tag{8}$$

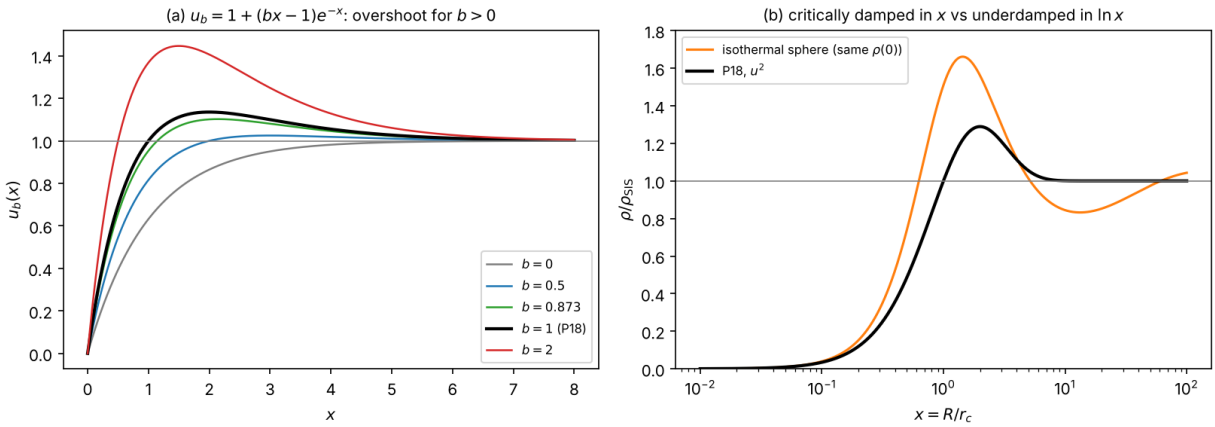


Figure 1: (a) The regular family $u_b = 1 + (bx-1)e^{-x}$: an overshoot for every $b > 0$, a positive mass excess for $b > 0.873$, P18 at $b = 1$. (b) Density relative to the SIS for P18 (black) and for the non-singular isothermal sphere with the same central density (orange): one critically damped overshoot against an underdamped oscillation in $\ln x$. Script: `fig_ufamily.py`.

so that $\rho/\rho_{\rm SIS} - 1 = -\delta \propto \xi^{-1/2}\cos(\frac{\sqrt{7}}{2}\ln\xi + \varphi)$. The isothermal sphere approaches the SIS *underdamped in* $\ln R$: its density oscillates about the SIS without end, with decreasing amplitude and no length scale. For the same central density ($r_c = \sqrt{8}\,r_0$) its first maximum is $\rho/\rho_{\rm SIS} = 1.66$ at $x = 1.44$ and its first minimum 0.83 at $x = 13$ (Fig. 1b). P18 approaches the SIS *critically damped in* R, with a single overshoot and the damping length r_c. A mechanism for P18 must therefore carry its own length scale; plain thermalisation, the classic idea behind isothermal cores, would give the scale-free ringing of the isothermal sphere.

5 A two-stage reading

The double root has a simple interpretation. Write $(D+1)^2 g = 0$ as the chain

$$(D+1)\,Q = 0, \qquad (D+1)\,g = Q. \tag{9}$$

The first gives $Q = Q_0 e^{-x}$: a quantity Q relaxes exponentially with the length r_c. The second says that g relaxes, with the same length, towards the value that Q imposes; its solution is $g = (g_0 + Q_0 x)\,e^{-x}$, the factor x coming from the resonance of the two equal rates. For P18, $g_0 = -1$ and $Q_0 = 1$. Critical damping is thus equivalent to two identical first-order relaxations in series, like two compartments with equal rate constants; unequal lengths $\lambda_1 \neq \lambda_2$ would give a sum of two exponentials (overdamping), and an oscillatory coupling would give ringing.

Remark. This is a reformulation, not a derivation. A physical theory of u would have to name the two processes, explain why they share the length r_c and why the central conditions select $b = 1$. Nothing in this note does so.

6 A burst experiment

6.1 The idea and the model

One physical reading of a single damped overshoot is an outflow slowed down by the infall: early central starbursts blow gas out, and the continuing accretion stops it. Dark matter is collisionless and feels the gas only through gravity, so the test must ask whether repeated changes of the central baryonic potential rearrange the dark matter in the P18 pattern: a deficit inside r_c, a single excess just outside it.

We use the one-dimensional collisionless shells of the paper (its Appendix H; units $G = v = 1$): 2000 shells with masses spaced logarithmically from 10^{-4} to 1, each following its cycloid with $E = -\frac{1}{2}$ until turnaround, where it receives $j = \sqrt{2}\lambda R_{\rm ta}$ ($\lambda = 0.035$), and afterwards

$$\ddot{r} = -\frac{M_{\rm enc}(r)}{r^2} + \frac{j^2}{r^3} - \frac{M_b(t)\, r}{[r^2 + a(t)^2]^{3/2}}, \tag{10}$$

with a softening $2 \times 10^{-4} t$, leapfrog integration with an adaptive global step, and shells free to cross. The last term is a prescribed central baryonic mass, a Plummer sphere of mass $M_b = f_b\, \xi_f\, t$ (a fraction f_b of the mass inside the front, $\xi_f = 0.2$) and radius $a(t)$ that jumps between a compact value $a_c\, t$ ($a_c = 0.002$, about a third of the model's core) and an expanded value $a_e\, t$. A cycle lasts $P\, t$ (uniformly spaced in $\ln t$), compact for half of it. All scales grow as t, so the runs stay self-similar. We stack 16 snapshots between 0.7 and 1 times the turnaround time of the outermost shell and use $\xi = R/vt$; the front is at $\xi \simeq 0.21$ and the paper's core at about $\xi_b/y \simeq 0.006$. The run without baryons is the reference.

Diagnostics. (a) The logarithmic slope of the dark-matter density at $\xi = 0.002$–0.006. (b) The displaced dark matter, $F(\xi) = M_{\rm ref}(<\xi) - M_{\rm run}(<\xi)$ (per unit t), the analogue of $x - m(x)$ for P18: positive where dark matter has been removed, negative where it has accumulated. The noise of F, estimated from the even minus the odd snapshots of the reference run, is at most 1.2×10^{-3}.

6.2 Results

Table 1: The runs (f_b: baryonic mass fraction; a_e: expanded radius; P: cycle length in units of t). Slope: dark-matter density at $\xi = 0.002$–0.006. F: displaced dark matter, its peak and the radius beyond which it turns negative (accumulation). Template rms: residual of the best fit of $A\,(x - m(x))$, $x = \xi/\xi_s$, in units of the peak, over $\xi = 0.003$–1.

run	f_b	a_e	P	slope	peak of F (at ξ)	$F < 0$ beyond	rms
reference	0	–	–	−2.46	–	–	–
fb05	0.05	0.05	0.05	−1.28	0.042 (0.20)	–	0.15
fb10	0.10	0.05	0.05	−0.77	0.069 (0.22)	–	0.13
fb20	0.20	0.05	0.05	−1.26	0.086 (0.22)	–	0.14
fb05fast	0.05	0.05	0.01	−1.02	0.051 (0.20)	–	0.16
fb10far	0.10	0.20	0.05	−1.44	0.072 (0.22)	–	0.12
fb05ae02	0.05	0.02	0.05	−1.17	0.035 (0.22)	0.61	0.16
fb05ae01	0.05	0.01	0.05	−1.80	0.020 (0.20)	0.48	0.19
fb02ae01	0.02	0.01	0.05	−2.27	0.006 (0.09)[a]	0.28[a]	0.26

[a] Only 3–5 times the noise.

1. *The bursts flatten the cusp.* The inner slope rises from −2.46 to between −0.8 and −1.8 (Table 1): repeated changes of the central potential raise the energy of the central dark matter, as required for the settling into a core and as found in hydrodynamical simulations (e.g. [1]). Weak confined bursts ($f_b = 0.02$) barely change the slope.

2. *The rearrangement has the P18 shape.* Scaled to its peak, the displaced mass rises, peaks and falls like $x - m(x)$ (Figs. 2c, 3b); with the gas confined, it even turns negative at 2.4–2.8 times the radius of the peak (P18: 3.3), a single zone of accumulation without ringing. The P18 template fits with a residual of 12–19% of the peak; a template without accumulation ($\arctan x$, the pseudo-isothermal sphere) fits worse in all runs butfb20, where the two are equal (14–28%).

3. *But on the wrong scale.* The peak lies at the front, $\xi \simeq 0.20$–0.22, about $35\, r_c$, in every run with a clear signal, and the accumulation lies beyond the front. Confining the gas to two or

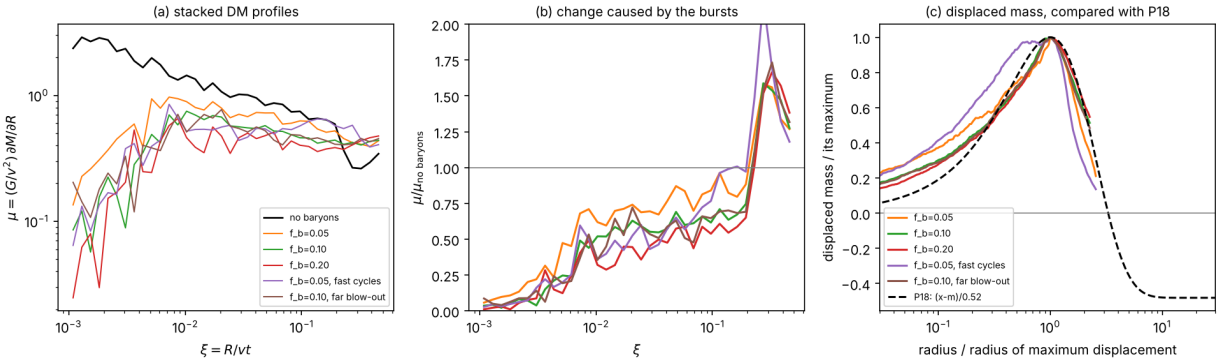


Figure 2: The first six runs. (a) Stacked dark-matter profiles, $\mu = (G/v^2)\,\partial M/\partial R$ against $\xi = R/vt$. (b) Ratio to the reference run: the dark matter is reduced at all radii inside the front and piles up just beyond it. (c) Displaced dark matter scaled to its peak, against radius scaled to the radius of the peak, compared with $(x - m)/0.52$ of P18. Script: `fig.py`.

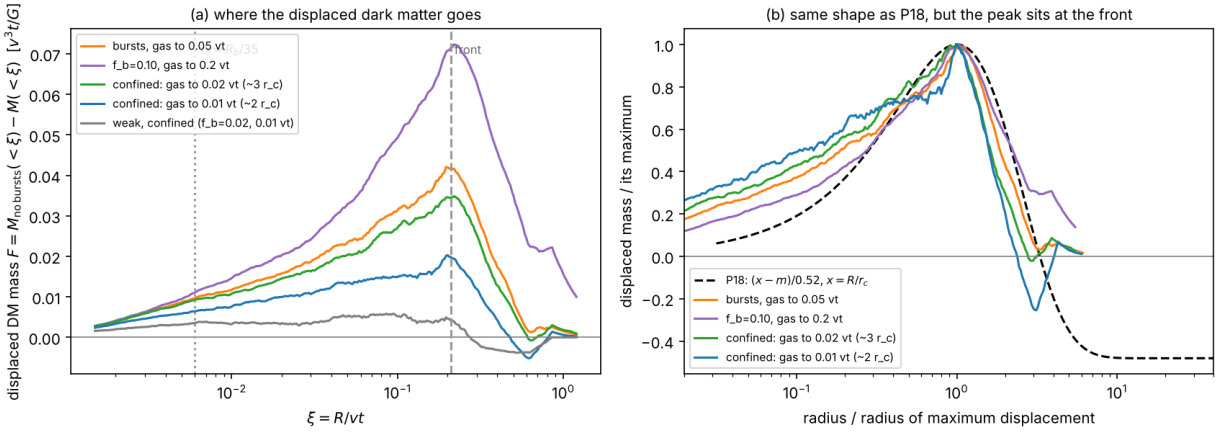


Figure 3: (a) Displaced dark matter for five runs, including the three with the gas confined to 0.01–0.02 vt, about 2–3 core radii; dotted: the model's core, dashed: the front. (b) The same, scaled to the peak, with P18. Script: `fig2.py`.

three core radii weakens the effect but does not move the peak inward (Fig. 3a). Inside the front the dark matter is reduced by 20–65% at all radii, so the flat curve is lost.

6.3 Why the matter goes to the front

In this one-dimensional model the dark matter moves on nearly radial orbits, which spend most of their time near their apocentres. A burst raises a particle's energy and moves its apocentre outward; the matter reappears far out, where the infall turns it round, i.e. at the front. Depositing it at about $2r_c$ would require orbits that spend their time near r_c, i.e. nearly isotropic orbits in the core. This is exactly the second requirement for the settling into a core found in the companion note *Halting and settling*: an energy lift *and* isotropy. The bursts supply the first, not the second. In three dimensions the second could come from the radial-orbit instability or a triaxial centre, which a spherical shell code cannot represent.

Caveats. The baryons are prescribed, not evolved, and switch instantly; the dark matter is spherical and one-dimensional; the cycling gas, 2–20% of the galaxy's mass, exceeds the 1.7% of baryons that the dwarfs keep today; and the reference run is a cusp, not P18. The results are indicative of a mechanism, not a model of real dwarfs.

7 Conclusions

1. The factor u of P18 is the unique solution of $u'' + 2u' + u = 1$, $u(0) = 0$, $u'(0) = 2$: a critically damped approach to the isothermal value, with the core radius as damping length (Proposition 1).

2. The regular solutions form the family $u_b = 1 + (bx - 1)e^{-x}$. A single overshoot exists for $b > 0$; a positive mass excess, $c(b) = b^2/4 + 3b/2 - 3/2$, for $b > 0.873$. P18 is $b = 1$, $c = \frac{1}{4}$, the member whose damping length equals the core radius (Proposition 2). Its density falls linearly at the centre.

3. The isothermal sphere approaches the SIS underdamped in $\ln R$, without a length scale; P18 critically damped in R, with the scale r_c.

4. Critical damping is equivalent to two identical exponential relaxations in series; their physical identity is open.

5. Bursts of central gas flatten the cusp and rearrange the dark matter in a P18-like pattern, but on the scale of the front, not the core. A single overshoot at about $2r_c$ needs, in addition to the energy, isotropic orbits in the core.

References

[1] Pontzen, A., & Governato, F. 2012, MNRAS, 421, 3464

Files in this package

`ode_checks.py`	numerical checks of Sects. 2–4; output `ode_checks_out.txt`
`fig_ufamily.py`	Fig. 1
`p18core.py`	P18 functions u, m (series near the centre)
`fbshells.py`	the shell code with the central baryonic component
`runall.py`	the nine runs (`python3 runall.py [names]`); logs `runs.log`, `runs2.log`
`run_*.pkl`	the stacked snapshots of the runs
`analyse.py`, `analyse2.py`	slopes, displaced mass, template fits; outputs `analyse_out.txt`, `analyse2_out.txt`
`fig.py`, `fig2.py`	Figs. 2 and 3 (run `analyse.py` on the first six runs before `fig.py`)